1 Lecture 09 Properties of OLS Short Answer Solutions

2 Question #1

Suppose

$$Y = X\beta + \boldsymbol{\varepsilon},$$

with

$$\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0.$$

The OLS estimator is

$$b = (X'X)^{-1}X'Y.$$

a. Show that the OLS estimator is conditionally unbiased:

$$\mathrm{E}(b \mid X) = \beta.$$

Solution:

See the lecture notes.

b. Using the result from part (a), show that the OLS estimator is unconditionally unbiased:

$$\mathrm{E}(b) = \beta.$$

Solution:

See the lecture notes.

3 Question #2

Suppose

$$Y = X\beta + \boldsymbol{\varepsilon}.$$

The OLS estimator is

$$b = (X'X)^{-1}X'Y.$$

a. Show that the OLS estimator can be written as

$$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}.$$

Solution:

Substituting $Y = X\beta + \boldsymbol{\varepsilon}$ into the OLS estimator,

$$\begin{aligned} b &= (X'X)^{-1}X'(X\beta + \boldsymbol{\varepsilon}) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}. \end{aligned}$$

Since

$$(X'X)^{-1}(X'X) = I,$$

it follows that

$$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}.$$

b. The conditional variance of the OLS estimator is

$$\mathrm{Var}(b \mid X) = \mathrm{E}\left[(b - \mathrm{E}(b \mid X))\,(b - \mathrm{E}(b \mid X))' \mid X\right].$$

Assuming that b is conditionally unbiased, show that

$$\mathrm{Var}(b \mid X) = (X'X)^{-1}X'\mathrm{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X\right] X(X'X)^{-1}.$$

Solution:

Using the result from part (a) and

$$\mathrm{E}(b \mid X) = \beta,$$

we have

$$\mathrm{Var}(b \mid X) = \mathrm{E}\left[\left((X'X)^{-1}X'\boldsymbol{\varepsilon}\right)\left((X'X)^{-1}X'\boldsymbol{\varepsilon}\right)' \mid X\right].$$

Using the properties of matrix transposes,

$$\left((X'X)^{-1}X'\boldsymbol{\varepsilon}\right)' = \boldsymbol{\varepsilon}'X(X'X)^{-1}.$$

Therefore,

$$\mathrm{Var}(b \mid X) = \mathrm{E}\left[(X'X)^{-1}X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X(X'X)^{-1} \mid X\right].$$

Conditioning on X treats all functions of X as fixed, so

$$\mathrm{Var}(b \mid X) = (X'X)^{-1}X'\mathrm{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X\right] X(X'X)^{-1}.$$

c. Assuming $\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0$, explain why

$$\mathrm{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X\right] = \mathrm{Var}(\boldsymbol{\varepsilon} \mid X).$$

Solution:

The conditional variance of $\boldsymbol{\varepsilon}$ is

$$\mathrm{Var}(\boldsymbol{\varepsilon} \mid X) = \mathrm{E}\left(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X\right) - \mathrm{E}(\boldsymbol{\varepsilon} \mid X)\mathrm{E}(\boldsymbol{\varepsilon} \mid X)'.$$

Since

$$\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0,$$

the second term is zero. Therefore,

$$\mathrm{Var}(\varepsilon \mid X) = \mathrm{E}\left(\varepsilon\varepsilon' \mid X\right).$$

d. Explain the implications of homoskedasticity and non-autocorrelation for the structure of $\mathrm{Var}(\boldsymbol{\varepsilon} \mid X)$. Show that under these assumptions,

$$\mathrm{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I_{n\times n},$$

where $\mathrm{Var}(\varepsilon_i \mid X) = \sigma^2$ and use this result to show that

$$\mathrm{Var}(b \mid X) = \sigma^2(X'X)^{-1},$$

Solution:

See the lecture notes.

4 Question #3

Suppose

$$Y = X\beta + \boldsymbol{\varepsilon}.$$

The OLS estimator can be written as

$$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}.$$

Let x_i' denote the ith row of X, so that x_i is a $K \times 1$ vector, and let ε_i denote the ith element of $\boldsymbol{\varepsilon}$. Define

$$w_i = x_i\varepsilon_i.$$

The sample average of w_i is

$$\bar{w} = \frac{1}{n}\sum_{i=1}^{n} w_i = \frac{1}{n}\sum_{i=1}^{n} x_i\varepsilon_i.$$

a. Show that the OLS estimator can be written as

$$b = \beta + \left(\frac{X'X}{n}\right)^{-1}\bar{w}.$$

Solution:

The matrix product $X'\boldsymbol{\varepsilon}$ can be written as

$$X'\boldsymbol{\varepsilon} = \begin{bmatrix} \sum_{i=1}^{n} x_{i1}\varepsilon_i \\ \sum_{i=1}^{n} x_{i2}\varepsilon_i \\ \vdots \\ \sum_{i=1}^{n} x_{iK}\varepsilon_i \end{bmatrix} = \sum_{i=1}^{n} x_i\varepsilon_i.$$

Therefore,

$$X'\boldsymbol{\varepsilon} = n\bar{w} = (n^{-1})^{-1}\bar{w}.$$

Substituting into the OLS estimator,

$$\begin{aligned} b &= \beta + (X'X)^{-1}(n^{-1})^{-1}\bar{w} \\ &= \beta + \left(\frac{X'X}{n}\right)^{-1}\bar{w}. \end{aligned}$$

b. Given that the OLS estimator is a function of the sample average $\bar{w}$ and that

$$\mathrm{E}(w_i) = 0,$$

explain what the Law of Large Numbers implies about the OLS estimator.

Solution:

See the lecture notes.

c. Given that the OLS estimator is a function of the sample average $\bar{w}$, explain what the Central Limit Theorem implies about the sampling distribution of the OLS estimator.

Solution:

See the lecture notes.